a) Develop a Java program to create an abstract class EnergySource with members Source ID, Source Name, and Energy Generated (in kWh). Include an abstract method calculateEfficiency(). Derive the classes SolarEnergy and WindEnergy from EnergySource and implement the method based on the following criteria: For SolarEnergy, efficiency is calculated as (Energy Generated / 5000) × 100. For WindEnergy, efficiency is calculated as (Energy Generated / 8000) × 100. Display source details and efficiency percentage. Demonstrate dynamic method dispatch using an EnergySource reference.

```
abstract class EnergySource {
    String sourceId;
    String sourceName;
    double energyGenerated;

    EnergySource(String sourceId, String sourceName, double energyGenerated) {
        this.sourceId = sourceId;
        this.sourceName = sourceName;
        this.energyGenerated = energyGenerated;
    }


    abstract double calculateEfficiency();


    void displayDetails() {
        System.out.println("Source ID: " + sourceId);
        System.out.println("Source Name: " + sourceName);
        System.out.println("Energy Generated: " + energyGenerated + " kWh");
        System.out.printf("Efficiency: %.2f\n", calculateEfficiency());
        System.out.println("1BM26CS483-T");
        System.out.println("CHANDANA BR");
        System.out.println("---------------------------------");
    }
}


class SolarEnergy extends EnergySource {

    SolarEnergy(String sourceId, String sourceName, double energyGenerated) {
        super(sourceId, sourceName, energyGenerated);
    }


    double calculateEfficiency() {
        return (energyGenerated / 5000.0) * 100.0;
    }
}
```

```java
class WindEnergy extends EnergySource {

    WindEnergy(String sourceId, String sourceName, double energyGenerated) {
        super(sourceId, sourceName, energyGenerated);
    }

    double calculateEfficiency() {
        return (energyGenerated / 8000.0) * 100.0;
    }
}


public class EnergyManagementSystem {
    public static void main(String[] args) {

        EnergySource ref;

        ref = new SolarEnergy("SOL-101", "Sahara Solar Park", 3750.0);
        ref.displayDetails();


        ref = new WindEnergy("WND-202", "Alta Wind Energy Center", 5600.0);
        ref.displayDetails();
    }
```

```
 ----jGRASP exec: java EnergyManagementSystem
Source ID: SOL-101
Source Name: Sahara Solar Park
Energy Generated: 3750.0 kWh
Efficiency: 75.00
1BM26CS483-T
CHANDANA BR
-----------------------------------
Source ID: WND-202
Source Name: Alta Wind Energy Center
Energy Generated: 5600.0 kWh
Efficiency: 70.00
1BM26CS483-T
CHANDANA BR
-----------------------------------

 ----jGRASP: Operation complete.
```